

LIGHTWEIGHT SAFE HELICOPTER (LSH)

DOCUMENT 00 — COMPREHENSIVE ENGINEERING BASELINE — TWIN ROTOR

Generation 1 V4.0 Variant — Twin Rotor

1. Status and Engineering Basis

Status: [PROPOSED] Preliminary controlled conceptual engineering baseline. This document is a variant study derived from the LSH V4.0 Gen1 baseline. Numerical values are targets, assumptions or preliminary calculations and are not demonstrated performance, certification results or flight-proven values.

The common LSH survival philosophy is retained: automatic detection, independent backup initiation, independent manual crew authority, deterministic Safety Kernel authorization, selective physical separation, occupant survival, recovery and post-event survival. Variant-specific hazards and failure modes must be analyzed separately.

The current LSH V4.0 baseline is a 20,000 kg-class single-main-rotor aircraft with a 21.0 m main rotor, 3.5 m tail rotor, 1,000 km range target, occupied survival capsule and eight fixed short supports. This variant changes the lift/propulsion architecture and therefore requires independent mass, CG, dynamics, structures, performance and certification closure.

2. Top-Level Technical Baseline

Parameter	Variant baseline	Status
Design / operating mass target	21,000 kg	TARGET
Architecture	Twin large-rotor tandem configuration; two longitudinal lifting rotors	CONCEPT
Main rotor diameter	24.0 m each	TARGET / STUDY
Total idealized lifting disk area	904.8 m ²	CALC
Weight-based total disk loading	23.2 kg/m ²	CALC
Tail rotor	None in baseline; yaw by differential rotor torque/control	CONCEPT
Propulsion	Two independent engine groups driving the two rotor systems	STUDY
Occupancy	10–12 persons	TARGET / STUDY
Survival capsule	Dedicated occupied final landing boundary	REQ
Landing supports	8 short, thick fixed mechanical supports under capsule	TARGET
Capsule mass target	5,400 kg	TARGET / STUDY
Range target	1,000 km	TARGET
Recovery	Distributed multi-canopy capsule recovery; preliminary 8-canopy architecture	STUDY
Retractable landing gear	None	LOCKED CONCEPT

3. Preliminary Engineering Calculations

Total idealized rotor disk area = $2 \times \pi \times (12.0 \text{ m})^2 = 904.8 \text{ m}^2$.

Weight = $21,000 \times 9.80665 = 205.9 \text{ kN}$.

Ideal sea-level hover induced power for the combined disks $\approx 1.98 \text{ MW}$.

Illustrative shaft power at figure of merit 0.75 $\approx 2.65 \text{ MW}$. This is not an installed-engine specification.

Actual tandem-rotor power must include profile power, interference, transmission losses, control margins, climb, cruise, hot/high and OEI requirements.

Two longitudinal rotor disks create strong coupling between rotor interference, longitudinal control, structural loads and emergency separation trajectories.

4. Multi-Layer Survival Architecture

L1. Primary automatic emergency detection and initiation.

L2. Independent physically/electrically segregated backup initiation.

L3. Independent manual crew authority.

L4. Deterministic Safety Kernel authorizes any hazardous actuation only after physical preconditions are satisfied.

- L5.** Selective staged separation of designated engine/fuel/drive/external hazards; rotor release is not baseline unless analysis proves it safe.
- L6.** Capsule structural survival with internal protective cell, energy-absorbing seats and eight fixed impact supports.
- L7.** Distributed capsule recovery system with independent emergency power and capsule-clear verification.
- L8.** Post-landing survival, locator, communications, rescue interface and immutable event record.

5. Variant-Specific Safety / Failure Cases

- Tandem rotor gearbox/shaft failure and loss of one rotor drive.
- Yaw/roll/pitch coupling following partial power loss.
- Rotor-to-rotor aerodynamic interference and transient control authority.
- Single-rotor or asymmetric emergency state must be modeled with full 6-DOF dynamics.
- Released hardware must have verified clearance from both rotors and the capsule.
- Fuel isolation and fire containment must be coordinated with separation.
- Structural loads at forward/aft rotor pylons and fuselage bending must be closed.
- Emergency parachute deployment must account for rotor wake and residual hardware.
- 8-support asymmetric touchdown and rollover stability.
- Common-cause fire, power, wiring, control-computer and structural damage.

6. Core System Requirements

- 01.** Aircraft shall maintain controllable flight following defined single-engine or single-drive failure cases where required by the selected certification basis.
- 02.** Each rotor system shall have independent power, control and monitoring architecture to the extent required by safety analysis.
- 03.** Yaw control shall not depend on a conventional tail rotor.
- 04.** Both rotor disks and their structural attachments shall be included in emergency clearance analysis.
- 05.** Partial/asymmetric release shall be explicitly sensed and modeled.
- 06.** Capsule separation shall be authorized only after validated clearance and recovery-envelope checks.
- 07.** Al/predictive functions may assist but shall not bypass the deterministic Safety Kernel.
- 08.** Human emergency authority shall remain non-bypassable.

7. Verification and Development Gates

G0 architecture freeze → G1 mass/CG/inertia and analytical closure → G2 subsystem demonstrations → G3 integrated ground demonstration → G4 6-DOF/dynamic simulation correlation → G5 structural/crash testing → G6 recovery-system testing → G7 flight development → G8 certification/qualification evidence.

Variant-specific evidence must include performance closure, structural loads, propulsion/rotor or lift-system analysis, emergency-state dynamics, selective separation trajectories, capsule clearance, recovery deployment, impact attenuation and common-cause/independence analysis.

8. IP Position

This variant may strengthen the breadth and commercial applicability of the LSH platform if the variants share a genuine common survival architecture while solving materially different vehicle-level problems. It does not, by itself, establish novelty, patentability, freedom-to-operate or market value. Those require prior-art and legal analysis.

9. Critical Open Items

- Tandem rotor gearbox/shaft failure and loss of one rotor drive. [OPEN]
- Yaw/roll/pitch coupling following partial power loss. [OPEN]
- Rotor-to-rotor aerodynamic interference and transient control authority. [OPEN]
- Single-rotor or asymmetric emergency state must be modeled with full 6-DOF dynamics. [OPEN]
- Released hardware must have verified clearance from both rotors and the capsule. [OPEN]
- Fuel isolation and fire containment must be coordinated with separation. [OPEN]
- Structural loads at forward/aft rotor pylons and fuselage bending must be closed. [OPEN]
- Emergency parachute deployment must account for rotor wake and residual hardware. [OPEN]
- 8-support asymmetric touchdown and rollover stability. [OPEN]

- Common-cause fire, power, wiring, control-computer and structural damage. [OPEN]

10. Engineering Integrity Note

No target in this document shall be presented as achieved. Detailed design requires aerodynamic/propulsive modelling, structural finite-element analysis, 6-DOF simulation, component testing, integrated ground testing and controlled flight testing as applicable.